

Chapter 157 — Diagnosis and Staging of Prostate Cancer — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Prostate-specific antigen (PSA) risk is continuous** — no level is risk-free; PSA 0–1 ng/mL → ~9% cancer (PCPT, Prostate Cancer Prevention Trial).
- **ERSPC (European Randomised Study of Screening for Prostate Cancer):** 20% ↓ cancer mortality at 16 yr; invite 570, diagnose 18 → 1 death prevented.
- **PLCO (Prostate, Lung, Colorectal and Ovarian trial):** no mortality benefit — contamination, low power.
- **Most important PSA determinant: prostate disease** (cancer, benign prostatic hyperplasia [BPH], prostatitis).
- **5 α -reductase inhibitor (5-ARI) $\geq$ 12 mo → PSA $\times$ 2; $\times$ 2.3 at 2 yr, $\times$ 2.5 at 7 yr.**
- **Repeat raised PSA before biopsy;** no antibiotics for PSA alone.
- **Free PSA percentage (%fPSA) 25%** at PSA 4–10 → ~95% cancers found, ~20% biopsies spared.
- **Seminal vesicle invasion = T3b;** non-regional nodes = **M1a**.
- **Grade group (GG) 2 = 3+4, GG3 = 4+3.**
- **Low/intermediate risk → no routine CT or bone scan;** high risk → bone scan + CT or multiparametric MRI (mpMRI).

DIAGNOSIS

- **Screening decision: shared** — USPSTF (U.S. Preventive Services Task Force) grade C 55–69; **against $\geq$ 70.**
- **USPSTF history:** 2008 against $\geq$ 75 → 2012 **grade D** → 2018 grade C.
- **AUA (American Urological Association):** 55–69, **2-yearly** interval.
- **NCCN (National Comprehensive Cancer Network):** baseline PSA at 45, repeat 1–4 yr.
- **American Cancer Society:** from 50; earlier if Black or first-degree relative <65.
- **Overtreatment worst >65 yr;** mean age at diagnosis ~66.
- **PSA era:** mortality ↓ >40%.
- **Malmö:** PSA in 40s predicts metastasis/death >25 yr later.
- **Biopsy thresholds:** 4 (original FDA) · 2.5–3 (common) · 3 (ERSPC) · 10 if >70 yr.
- **Digital rectal examination (DRE):** little value at PSA <3; most urologists **combine PSA + DRE.**
- **Suspicious DRE $\approx$ doubles biopsy yield:** PSA 4–10 → 17–32% vs 45–51%; >10 → 43–65% vs 70–90%.
- **PSA $\uparrow$:** volume $\gg$ age; Black ethnicity; biopsy, massage, transurethral resection.
- **PSA $\downarrow$:** hypogonadism, obesity, statins, 5-ARI, BPH surgery.
- **DRE → no false-positive PSA** (within assay error).
- **Ejaculation:** repeat after **48 h abstinence** if minimally raised.
- **Finasteride 1 mg = 5 mg** for PSA lowering.
- **5-ARI:** baseline PSA first; alternatively on-treatment **nadir** = new baseline.
- **PSA density (PSAD) $\geq$ 0.15** → biopsy (PSA 4–10, normal DRE); also active surveillance eligibility.
- **PSA velocity (PSAV) >0.75 ng/mL/yr** (PSA 4–10); **0.4** at lower PSA.
- **PSAV >0.35** 10–15 yr pre-diagnosis → lethal cancer; >2 in year before → cancer death after treatment.
- **%fPSA 25% $\approx$ PSAD 0.078** in specificity, no volume needed.
- **%fPSA unaffected** by ethnicity or 5-ARI.
- **Complexed PSA (cPSA):** specificity > total PSA, $\approx$ %fPSA.
- **proPSA → cancer; benign PSA (bPSA) → BPH;** human kallikrein 2 (hK2) tracks grade.
- **Prostate Health Index (phi) = tPSA + fPSA + [–2]proPSA;** FDA-approved.
- **4Kscore = tPSA, fPSA, intact PSA, hK2.**

- **Prostate cancer gene 3 (PCA3), TMPRSS2:ERG:** urine; aggressiveness data conflicting.
- **ConfirmMDx:** methylation of **benign biopsy tissue** → avoids repeat biopsy.
- **tPSA alone rarely triggers biopsy** → add biomarkers + clinical risk.

STAGING

- **Pathological > clinical stage** for accuracy and outcome prediction.
- **Post-prostatectomy predictors:** grade, margins, extracapsular extension, seminal vesicle invasion, nodes.
- **TNM:** adopted **1975;** current **2017;** Whitmore–Jewett historical.
- **T1a $\leq$ 5% · T1b >5%** resected tissue; **T1c** needle-biopsy detected.
- **T2a $\leq$ ½ one side · T2b >½ one side · T2c both sides.**
- **T4:** fixed or invades sphincter, rectum, bladder, levator, pelvic wall.
- **M1a nodes · M1b bone · M1c other $\pm$ bone.**
- **Organ-confined:** ~80% PSA <4 · ~66% 4–10 · <50% >10.
- **Node-positive:** ~20% PSA >20 · ~75% PSA >50.
- **High-grade tumours → less PSA per gram.**
- **Prostatic acid phosphatase:** obsolete.
- **Most important biopsy finding: grade;** Gleason most used.
- **Gleason = predominant + second pattern, 2–10;** tertiary 5 upgrades (2005).
- **Grade groups:** 1 $\leq$ 6 · 2 3+4 · 3 4+3 · 4 = 8 · 5 = 9–10.
- **Cribriform/intraductal, % positive cores** → prognosis, active surveillance selection.
- **D'Amico low:** T1–2a + PSA $\leq$ 10 + Gleason $\leq$ 6 → 83% disease-free at 10 yr.
- **D'Amico intermediate:** T2b or PSA >10–<20 or Gleason 7 → 46%.
- **D'Amico high:** T2c or PSA >20 or Gleason 8–10 → 29%.
- **MRI before repeat biopsy after negative biopsy;** key points: before any biopsy.
- **Suspicious MRI lesion → targeted biopsy** (MRI–transrectal ultrasound fusion).
- **mpMRI = T2 + diffusion-weighted + dynamic contrast-enhanced.**
- **PSMA (prostate-specific membrane antigen) PET:** ^{68}Ga -PSMA-11, ^{18}F -DCFPyL — approved for **initial high-risk staging.**

- **Genomic tests:** Oncotype DX **17 genes** · Decipher **22 RNA** · Prolaris cell-cycle.
- **Pelvic lymph node dissection (PLND):** no consistent oncological benefit → **nomogram-guided, staging only.**

CLASSIC TRAPS

- **Trap:** normal PSA excludes cancer — **Actually:** ~9% at PSA 0–1.
- **Trap:** PLCO proves screening futile — **Actually:** contaminated control arm.
- **Trap:** seminal vesicle invasion is T4 — **Actually:** T3b.

- **Trap:** non-regional nodes are N1 — **Actually:** M1a.
- **Trap:** bilateral impalpable cancer is T2c — **Actually:** T1c.
- **Trap:** 3+4 and 4+3 same group — **Actually:** GG2 vs GG3.
- **Trap:** 5-ARI lowers %fPSA — **Actually:** tPSA halves, %fPSA unchanged.
- **Trap:** fPSA higher in cancer — **Actually:** lower.
- **Trap:** D'Amico low risk needs one criterion — **Actually:** all three.
- **Trap:** stage low-risk with bone scan — **Actually:** not routinely.